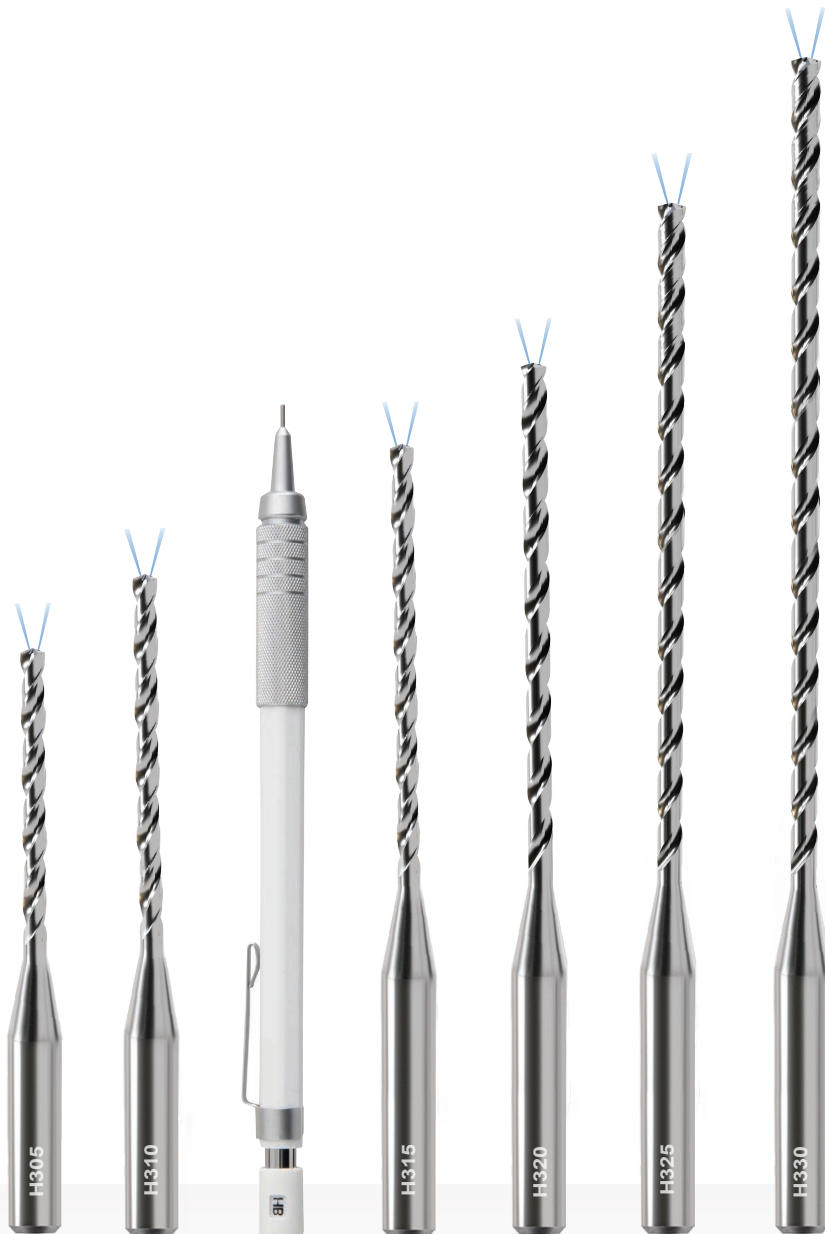


INDRILL Cool

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30



**High Precision
Deep Hole Drills**

Ø 0.80 - Ø 6.00 mm

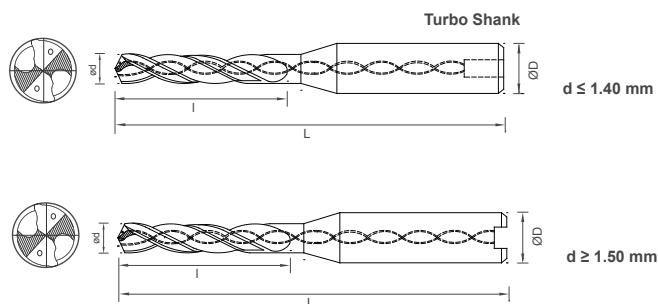
l / d
5x
10x
15x
20x
25x
30x

High Precision Deep Hole Drills



APPLICATION

Specially designed for deep hole Drilling in unalloyed and alloyed Steels, Stainless Steel, Cast Iron and heat treated Steels upto 50 HRC



Dia h6 ød	Shank h6 ØD		H305 5xd		H310 10xd		H315 15xd		H320 20xd		H325 25xd		H330 30xd	
			I	L	I	L	I	L	I	L	I	L	I	L
0.80			8.00	45.00	12.00	55.00	16.00	55.00	20.00	65.00				
0.90			8.00	45.00	12.00	55.00	16.00	55.00	20.00	65.00				
1.00			10.00	45.00	16.00	55.00	18.00	75.00	25.00	75.00	30.00	75.00	35.00	75.00
1.10			10.00	45.00	16.00	55.00	20.00	75.00	25.00	75.00	30.00	75.00	35.00	75.00
1.20			10.00	45.00	16.00	55.00	24.00	75.00	28.00	75.00	35.00	75.00	45.00	85.00
1.30			12.00	55.00	18.00	55.00	24.00	75.00	30.00	75.00	35.00	85.00	45.00	85.00
1.40			12.00	55.00	18.00	55.00	27.00	75.00	32.00	75.00	40.00	85.00	45.00	85.00
1.50			12.00	55.00	18.00	55.00	27.00	75.00	36.00	80.00	45.00	85.00	55.00	95.00
1.60			16.00	60.00	26.00	65.00	30.00	75.00	36.00	80.00	45.00	85.00	55.00	95.00
1.70			16.00	60.00	26.00	65.00	30.00	75.00	40.00	85.00	50.00	95.00	55.00	95.00
1.80			16.00	60.00	26.00	65.00	35.00	75.00	40.00	85.00	50.00	95.00	65.00	105.0
1.90			16.00	60.00	26.00	65.00	35.00	75.00	45.00	95.00	55.00	95.00	65.00	105.0
2.00	4.00		16.00	60.00	26.00	65.00	35.00	80.00	45.00	95.00	55.00	95.00	65.00	105.0
2.10	4.00		18.00	60.00	33.00	75.00	40.00	80.00	50.00	95.00	60.00	108.0	75.00	115.0
2.20	4.00		18.00	60.00	33.00	75.00	40.00	80.00	50.00	95.00	60.00	108.0	75.00	115.0
2.30	4.00		20.00	60.00	33.00	75.00	40.00	80.00	55.00	95.00	65.00	108.0	75.00	115.0
2.40	4.00		20.00	60.00	33.00	75.00	45.00	90.00	55.00	95.00	70.00	112.0	85.00	130.0
2.50	4.00		20.00	60.00	33.00	75.00	45.00	90.00	60.00	108.0	70.00	112.0	85.00	130.0
2.60	4.00		22.00	60.00	40.00	80.00	48.00	90.00	60.00	108.0	75.00	120.0	85.00	130.0
2.70	4.00		22.00	60.00	40.00	80.00	48.00	90.00	65.00	108.0	75.00	120.0	95.00	140.0
2.80	4.00		24.00	65.00	40.00	80.00	50.00	90.00	65.00	108.0	80.00	125.0	95.00	140.0
2.90	4.00		24.00	65.00	40.00	80.00	50.00	90.00	68.00	112.0	80.00	125.0	95.00	140.0
3.00	5-10xd 4.00	15-30xd 6.00	24.00	65.00	40.00	80.00	55.00	100.0	68.00	112.0	85.00	130.0	105.0	150.0

Dia h6 ød	Shank h6 ØD		H305 5xd		H310 10xd		H315 15xd		H320 20xd		H325 25xd		H330 30xd	
			I	L	I	L	I	L	I	L	I	L	I	L
3.10	5-10xd 4.00	15-30xd 6.00	26.00	75.00	42.00	80.00	55.00	100.0	75.00	120.0	85.00	130.0	105.0	150.0
3.20	4.00	6.00	26.00	75.00	42.00	80.00	60.00	100.0	75.00	120.0	95.00	140.0	105.0	150.0
3.30	4.00	6.00	28.00	75.00	45.00	90.00	60.00	100.0	75.00	120.0	95.00	140.0	115.0	160.0
3.40	4.00	6.00	28.00	75.00	45.00	90.00	60.00	100.0	80.00	130.0	95.00	140.0	115.0	160.0
3.50	4.00	6.00	28.00	75.00	45.00	90.00	65.00	108.0	80.00	130.0	100.0	150.0	115.0	160.0
3.60	4.00	6.00	36.00	75.00	52.00	90.00	65.00	108.0	85.00	130.0	100.0	150.0	125.0	175.0
3.70	4.00	6.00	36.00	75.00	52.00	90.00	70.00	108.0	85.00	130.0	105.0	150.0	125.0	175.0
3.80	4.00	6.00	36.00	75.00	52.00	90.00	70.00	108.0	90.00	140.0	105.0	150.0	125.0	175.0
3.90	4.00	6.00	36.00	75.00	52.00	90.00	70.00	108.0	90.00	140.0	110.0	160.0	125.0	175.0
4.00	6.00		36.00	75.00	60.00	100.0	75.00	120.0	95.00	140.0	110.0	160.0	135.0	185.0
4.10	6.00		36.00	75.00	60.00	100.0	75.00	120.0	95.00	140.0	120.0	170.0	135.0	185.0
4.20	6.00		36.00	75.00	60.00	100.0	75.00	120.0	95.00	140.0	120.0	170.0	135.0	185.0
4.30	6.00		36.00	75.00	60.00	100.0	80.00	120.0	100.0	150.0	120.0	170.0	145.0	195.0
4.40	6.00		36.00	75.00	60.00	100.0	80.00	120.0	100.0	150.0	120.0	170.0	145.0	195.0
4.50	6.00		36.00	75.00	60.00	100.0	80.00	120.0	110.0	160.0	130.0	180.0	145.0	195.0
4.60	6.00		44.00	80.00	65.00	108.0	85.00	125.0	110.0	160.0	130.0	180.0	155.0	205.0
4.70	6.00		44.00	80.00	65.00	108.0	85.00	125.0	110.0	160.0	130.0	180.0	155.0	205.0
4.80	6.00		44.00	80.00	65.00	108.0	90.00	140.0	115.0	160.0	140.0	190.0	165.0	210.0
4.90	6.00		44.00	80.00	65.00	108.0	90.00	140.0	115.0	160.0	140.0	190.0	165.0	210.0
5.00	6.00		44.00	80.00	65.00	108.0	90.00	140.0	115.0	160.0	140.0	190.0	165.0	210.0
5.10	6.00		44.00	80.00	72.00	108.0	95.00	140.0	125.0	165.0	150.0	200.0	175.0	220.0
5.20	6.00		44.00	80.00	72.00	108.0	95.00	140.0	125.0	165.0	150.0	200.0	175.0	220.0
5.30	6.00		44.00	80.00	72.00	108.0	95.00	140.0	125.0	165.0	150.0	200.0	175.0	220.0
5.40	6.00		44.00	80.00	72.00	108.0	95.00	140.0	125.0	165.0	150.0	200.0	175.0	220.0
5.50	6.00		44.00	80.00	72.00	108.0	100.0	150.0	140.0	185.0	160.0	210.0	190.0	240.0
5.60	6.00		48.00	80.00	72.00	108.0	100.0	150.0	140.0	185.0	160.0	210.0	190.0	240.0
5.70	6.00		48.00	80.00	78.00	120.0	100.0	150.0	140.0	185.0	160.0	210.0	190.0	240.0
5.80	6.00		48.00	80.00	78.00	120.0	100.0	150.0	140.0	185.0	160.0	210.0	190.0	240.0
5.90	6.00		48.00	80.00	78.00	120.0	100.0	150.0	140.0	185.0	170.0	220.0	190.0	240.0
6.00	6.00		48.00	80.00	78.00	120.0	100.0	150.0	140.0	185.0	170.0	220.0	190.0	240.0

ON REQUEST :
Inch Sizes / Special sizes

PACKING :
Single tool per pack

Salient Features

Cutting Edge Stabilization

Very consistent mICRO geometry on cutting edge enables narrow process capability

Flute Geometry

Specially designed flute geometry with optimal core design for effective chip breaking and tool rigidity

Super finishing

Super finish flute for highly efficient chip evacuation

Unique Pre+Post Coat Treatment

Unique, specially developed pre and post Coating treatment leads to reduced heat generation and smooth chip flow providing excellent Tool life

Shank

Robust Shank with optimized neck provides for rigid and precise drilling for high L/D ratios

Turbo Chamber

Shank with Turbo Chamber for small Diameters up to 1.4mm ensures continuous coolant flow even at low pressure



Operational Guidance

- Tools for mICRO mACHINING are very delicate in nature and need very careful handling while storage, mounting, usage and dismounting.
- Avoid touching mICRO Tools with bare hands, it can cause injury.
- When removing tools from original packing tool case, be careful to not touch cutting edges.
- Please check tool dimensions and application material before starting operation.
- Ensure tool is firmly held by Collets / Tool holders with runout <0.003 mm.
- Tabulated parameters provide guidelines which are starting points for optimising speed and feeds at user end. Please ensure to adjust condition according to application and machining condition.
- Use suitable coolant as applicable to work material / machining environment. Coolant should be copious, continuous and exactly directed at cutting edges.
- Rigidity of elements : Tool holding, job holding and machine is very critical for optimum performance.
- Recommended coolant pressure ≥ 50 bar, at least 15 bar pressure is required.
- Use coolant filter with mesh $< 3\mu$ for coolant to prevent jamming inside coolant holes.

Cutting Performance

Pilot Drill

Dia	3.00 m7
Dilling Depth	9
Cutting Speed	52
feed	0.07

Pilot Drill

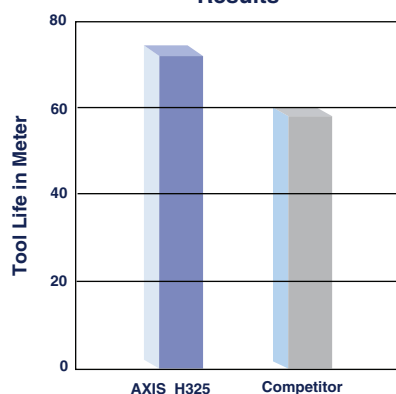
Dia	6.00 m7
Dilling Depth	18
Cutting Speed	66
feed	0.114

AXIS

Competitor

Version	H325 - Ø3x85mm	25D - Ø3mm
Material	C45	C45
Cutting Speed	79	70
Feed	0.09	0.08
Coolant	Water Soluble	Water Soluble
Coolant Pressure	15 bar	15 bar
Tool Life	75	60
Wear	Normal Wear	High wear marks

Results

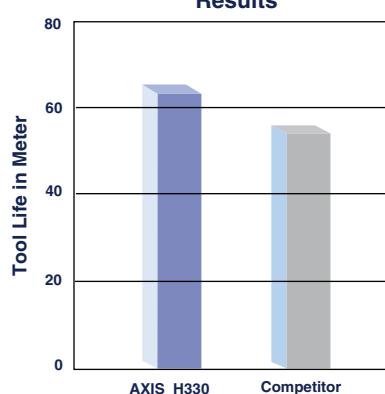


AXIS

Competitor

Version	H330 - Ø6x190mm	30D - Ø6mm
Material	C45	C45
Cutting Speed	80	70
Feed	0.2	0.18
Coolant	Water Soluble	Water Soluble
Coolant Pressure	15 bar	15 bar
Tool Life	65	55
Wear	Normal Wear	High wear marks

Results



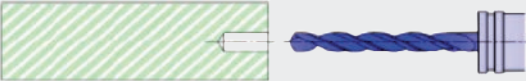
Excellent chip evacuation with short and curvy chips

H305 / H310 / H315 / H320 / H325 / H330

Materials					
			0.80-0.90	1.00-1.20	1.30-1.50
P	Low/Unalloyed Steel <600 N/mm²	N min-1	12500-16000	15300-19700	12300-15100
		F mm/rev	0.015-0.018	0.015-0.020	0.020-0.025
	Low/Unalloyed Steel 600-1000 N/mm²	N min-1	11500-15000	13700-17800	11000-13700
		F mm/rev	0.015-0.018	0.015-0.020	0.020-0.025
	High Alloyed Steel 700-1500 N/mm²	N min-1	10000-12000	10600-14000	8400-10800
		F mm/rev	0.012-0.015	0.012-0.015	0.016-0.022
M	Ferritic / Austenitic <700 N/mm²	N min-1	7100-9000	7900-10800	6300-8300
		F mm/rev	0.012-0.014	0.012-0.014	0.018-0.022
K	GCI/NCI <180 BHN	N min-1	12500-16000	15300-19700	12300-15100
		F mm/rev	0.015-0.018	0.015-0.020	0.020-0.025
	GCI/NCI 200-250 BHN	N min-1	10000-12300	11600-15200	9300-11700
		F mm/rev	0.012-0.015	0.012-0.015	0.015-0.020

Drilling Concept / Strategy

Step 1: Pilot Hole



- Uneven or Slant Surface must be spot faced before using Pilot drill
- Centering is essential during Horizontal machining
- Use AXIS Pilot drill to drill hole of minimum 2-3D
- Should you use third party pilot drill make sure the pilot drill dia. is minimum 10µ oversize & Point angle of Pilot drill is 140°
- Any chips from Pilot hole must be removed prior to drilling deep hole

Step 3: Deep Hole Drilling



- Start drilling the hole with recommended Cutting Data with continuous feed (No pecking up to 30D)
- Incase of Interrupted drilling/through holes reduce the feed by 50% prior to break out

H305 / H310 / H315	Parameters as per above	H325	Reduce rpm by 10% from above, keep feed same
H320	Reduce rpm by 5% from above, keep feed same	H330	Reduce rpm by 15% from above, keep feed same

Dia Range					
1.60-2.00	2.10-2.40	2.50-3.00	3.10-4.00	4.10-5.00	5.10-6.00
9200-12300	8600-10600	6900-8900	6600-9000	5700-7300	4700-5900
0.045-0.050	0.060-0.070	0.080-0.090	0.100-0.120	0.120-0.160	0.160-0.200
8200-11100	8200-10000	6500-8400	6200-8400	5400-7000	4500-5600
0.045-0.050	0.060-0.070	0.080-0.090	0.100-0.120	0.120-0.150	0.150-0.190
6300-8800	5900-7600	4700-6600	4600-6600	4100-5400	3400-4300
0.030-0.040	0.040-0.050	0.070-0.080	0.080-0.100	0.100-0.120	0.120-0.150
4700-6700	4600-6000	3700-5000	3500-5100	3100-4200	2600-3400
0.030-0.040	0.040-0.050	0.050-0.060	0.060-0.070	0.070-0.080	0.080-0.100
9200-12300	8600-10600	6800-8900	6600-9000	5700-7300	4700-5900
0.045-0.050	0.060-0.070	0.080-0.090	0.100-0.120	0.120-0.160	0.160-0.200
7000-9500	6600-8300	5300-7000	5100-7100	4500-5900	3800-4700
0.030-0.040	0.040-0.060	0.070-0.080	0.090-0.10	0.100-0.130	0.140-0.160

Step 2 : Entering Deep hole drill into Pilot Hole



- Internal Coolant must be turned off
- Set Max rpm of 500-1000 (preferably anti-clockwise), feed = drilling feed (mm/rev)
- Stop the drill 1-1.5mm before the depth of Pilot Hole

Step 4: Retract the Drill



- After reaching the maximum drilling depth, retract the drill by 1-2mm for chip removal
- Then reduce the rotation speed to 500-1000 rpm and retract the drill at feed of 500-700 mm/min (not rapid traverse)
- Internal coolant can be turned off in the area of pilot hole

INDRILL Cool



AXIS
Tools for mICRO mACHINING



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PEOPLE